Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Google Workspace Masterclass

Course Code: C148

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	Initial release of the Google Workspace Masterclass courseware: three topics, 25 self-contained Activities, current Google mechanics and native charts.	Tertiary Infotech Academy

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

How to use this guide....5

Before you start....5

Learning outcomes....5

Governance and Drive content lifecycle....5

Activity 1 Content policy and identity boundaries....5

Activity 2 Drive ownership and team structure....6

Activity 3 Sharing and effective access audit....8

Activity 4 Posting lifecycle and asset metadata....9

Activity 5 Search recovery and version diagnosis....11

Content assets and customer web properties....13

Activity 6 Docs structure and collaborative review....13

Activity 7 Docs templates and publication readiness....14

Activity 8 Sheets import schema and data validation....15

Activity 9 Sheets formulas and operational checks....17

Activity 10 Filter views pivots and editable charts....18

Activity 11 Slides assets and customer presentation....19

Activity 12 Forms publication consent and response schema....21

Activity 13 Forms validation and customer feedback analysis....22

Activity 14 Gmail routing labels and search controls....23

Activity 15 Gmail delegation and customer response governance....25

Activity 16 Calendar availability and booking boundaries....26

Activity 17 Meet collaboration and recording controls....28

Activity 18 Chat spaces tasks and whiteboarding replacement....29

Activity 19 Sites architecture navigation and draft content....30

Activity 20 Sites publishing embed access and maintenance....32

Activity 21 CMS market offerings and weighted selection....33

Metrics add-ons and integration decisions....35

Activity 22 Workspace metrics and evidence provenance....35

Activity 23 Visitor journey funnel and improvement tests....36

Activity 24 Add-on selection scopes and administrator controls....37

Activity 25 Apps Script and API integration contract....39

Supplementary mechanisms and legacy coverage....40

Domain verification and mail routing....40

User alias group and access contract....41

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Suspension handover and deletion gate....42
Service availability and account security controls....42
Workspace content tools and boundaries....43
Communication and optional extension selection....44
Retired applications and current replacements....44
Meet Companion mode and echo prevention....45
Relative absolute and named references....45
Rounding and conditional averages....46
Extremes ranking and ties....47
Conditional minimum maximum and no matches....47
Exact match lookup and source identity....48
Slicers filters and formula visibility....49
Quick reference....50
Support....50

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

How to use this guide

This guide explains how Harbour Home Supplies, a fictional SME, controls content and access, creates and maintains customer assets, and uses defined metrics and reviewed integrations. Each Activity includes complete steps, mock records, acceptance checks and troubleshooting. Procedures appear here and in activity folders; the slide deck teaches mechanisms and evidence.

Before you start

Required: current browser, training Google Account, Drive, Docs, Sheets, Slides, Forms and Sites. Calendar/Meet/Chat, shared drives, Admin reports, appointment schedules and extensions depend on role, edition and administrator policy. Practise only in new training containers. When unavailable, complete the documented fallback and record the evidence gap rather than claim the feature was executed.

Learning outcomes

Outcome	Capability
LO1	Implement content management policies and post and update content using Google Drive.
LO2	Maintain web properties and assets, create and curate content, and maintain awareness of market offerings.
LO3	Report performance metrics, recommend appropriate add-ons, and generate ideas to improve customer and visitor experience.

Governance and Drive content lifecycle

LO1 · Slide range 10–55

Activity 1 Content policy and identity boundaries

20 minutes · Slides 11–19

Harbour Home Supplies is a fictional Singapore SME with Sales, Operations and Finance teams. Its migration contains supplier invoices, customer enquiries and public product sheets. An external agency needs only approved marketing assets.

Goal: Completed policy covers classification, ownership, posting, review and exceptions; access evidence names the tested account and role.

Tools: Drive, Docs; administrator demonstration

Additional mechanism/chart slides: 238,239,240,241.

Mock inputs: classification-register.csv, policy-template.md, scenario.json.

Local package: activities/activity-01-content-policy-and-identity-boundaries/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Classification	Internal / Confidential / Public	Policy field, not automatic enforcement
Owner	Finance lead	Accountable for review
General access	Restricted	Default for internal records
Review date	2026-10-01	Monthly permission review

Policy versus setting: A policy assigns accountability; Drive sharing is the enforcement action. A document describing Restricted does not change any permission.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Identity boundary: An email alias is not a separate account. Groups grant access to their current members; membership therefore needs review.

Administrator boundary: Only an authorised administrator changes domain or service controls. Learners draft a change request rather than alter a live tenant.

Public release: Copy only approved non-sensitive content to the public asset area. Keep invoices in an independently restricted container.

Detailed procedure

1. Open the local policy-template.md and classification-register.csv. Identify which example assets contain confidential data.

2. In Drive create a new training folder named HHS Activity 01 followed by your initials. Keep General access Restricted.

3. Create a Google Doc named HHS Content Policy v 1 and paste the policy template. Complete owner, classification, access, publication, retention and exception sections.

4. For each register row assign an accountable owner, permitted audience, update frequency and review date. Use Finance-only access for invoices.

5. Share only the training policy document with your paired learner as Commenter. Do not use Anyone with the link.

6. Ask your partner to open the document with the invited account and record their role. Remove the training grant after the test.

7. Draft an administrator change request describing external-sharing boundaries and evidence needed. Do not change a production domain.

8. Export the policy as PDF into your output folder and complete the evidence checklist with the tested identity and observed outcome.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
POL-001	Policy approved	Named owner and monthly review
ACL-001	Internal folder	Restricted; staff group only
TEST-001	Agency account	Invoice access denied
EXC-001	Agency product draft	Commenter; approval recorded
Symptom	Cause	Corrective check
Policy says restricted; link opens	General access is broader than the written policy	Restrict actual sharing and test another identity
Group includes former staff	Membership outlasted employment	Remove group membership; review direct grants
No administrator rights	Tenant settings inaccessible	Produce administrator request and evidence plan

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 2 Drive ownership and team structure

20 minutes · Slides 20–28

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

A product catalogue must remain with Harbour Home Supplies when its editor leaves. A working draft currently sits in an individual My Drive folder. Decide where the source of record should live.

Goal: Folder structure and ownership register distinguish individual from organisation ownership; least-privilege role and unavailable-feature fallback are recorded.

Tools: Drive; shared drive if available

Additional mechanism/chart slides: None.

Mock inputs: ownership-register.csv, product-catalogue.csv, scenario.json.

Local package: activities/activity-02-drive-ownership-and-team-structure/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
My Drive owner	Individual account	Lifecycle includes ownership handover
Shared drive owner	Organisation	Files remain when a member leaves
Shared drive role	Manager / Content Manager / Contributor	Use least privilege
Folder exception	Limited access	Available to authorised managers

Persistence: Organisation ownership is the reason to use shared drives for team records; it is separate from the ability to edit.

Role selection: Managers manage membership and settings. Contributors add and edit content without the broader Manager powers.

Move boundary: Moving an existing file can change effective access. Inspect direct grants and drive restrictions before and after a move.

Availability: Shared drive availability depends on account and administrator settings. A Restricted My Drive training folder is the fallback, with a documented handover risk.

Detailed procedure

1. Read ownership-register.csv and draw the current author-to-file relationship.

2. If your training account has Shared drives and permission to create one, create HHS Marketing Training. Otherwise create a Restricted My Drive folder and note the fallback.

3. Create Drafts, Approved Assets and Archive subfolders. Do not add confidential data to a broadly accessible drive.

4. Upload product-catalogue.csv into Drafts and create an accompanying catalogue Google Doc.

5. Inspect the item location and Share dialog. Record who owns the content and which role your account holds.

6. Ask an authorised trainer to demonstrate member roles where available; assign your partner only the minimum permitted role for the exercise.

7. Record the expected outcome if the editor leaves and identify the administrator or owner responsible for the handover.

8. Capture the folder structure and ownership/role evidence; remove temporary grants after verification.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Acceptance and troubleshooting

Evidence	Artifact	Expected result
OWN-001	CAT-001 location	Shared drive or documented fallback
ROLE-001	Editor identity	Contributor, not Manager
EXIT-001	Member removed	Organisation source remains
VIEW-001	Agency identity	Only approved asset available
Symptom	Cause	Corrective check
Shared drives not visible	Account or policy does not offer access	Use fallback; record limitation
Agency sees too much	Broad drive membership	Use narrow approved sharing or separate release area
Move changes access	Restrictions differ between containers	Compare effective audience before and after

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 3 Sharing and effective access audit

25 minutes · Slides 29–37

Finance deleted an agency user from one invoice file, but the agency still opens it through a group or public link. Diagnose all access paths rather than judging the direct People list alone.

Goal: Access register identifies all four grant paths; one approved allow and one uninvited deny outcome are recorded for the exact training asset.

Tools: Drive, two test identities

Additional mechanism/chart slides: None.

Mock inputs: access-notes.md, access-register.csv, mock-invoice.csv, scenario.json.

Local package: activities/activity-03-sharing-and-effective-access-audit/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
People with access	Account or group	Inspect direct and inherited paths
General access	Restricted / audience	May grant access without individual entry
Limited access	Folder-level control	Cannot manually apply to one file
Manager / owner	Privileged principal	Cannot remove their underlying access

Union of grants: Removing one grant does not remove other access paths. A group or general audience can still authorise the account.

Inheritance: Adjust inherited sharing at its origin, or use an appropriate limited-access folder. Do not assume a child removal overrides all parent access.

Limited-access exception: Current Drive supports folder-level limited access. Shared drive Managers and My Drive owners retain their privileged access.

Verification: A Share dialog is configuration evidence. A separate account opening or being denied the exact file is outcome evidence.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Detailed procedure

1. Open access-register.csv and mark direct, group, inherited and general grants for each asset.

2. Create only synthetic invoice and product files in new training folders. Use the provided mock-invoice.csv.

3. Share a harmless product file with your partner, then inspect People with access and General access separately.

4. Remove the partner direct grant and ask the partner to test the same URL. Record the actual outcome.

5. If your role offers folder Limited access, demonstrate it only on a training subfolder. If unavailable, record the role/policy limitation and use a separate Restricted folder.

6. Correct any broad training grants at their actual origin. Do not remove permissions on existing business folders.

7. Test the exact file as the approved identity and as an uninvited identity. Record allow and deny results without exposing real accounts publicly.

8. Complete the audit findings and corrective actions, then remove temporary sharing.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
ACL-BEFORE	Agency	Opens through broader grant
ACL-AFTER	Agency	Denied after full correction
FIN-ALLOW	Finance tester	Can read expected file
REVIEW	Owner	Signs effective-access register
Symptom	Cause	Corrective check
Removed user still opens	Group / inherited / general grant remains	Inspect and correct each authorised path
Folder limited; public file opens	File has another general/direct grant	Review both General access and People
Cannot modify inherited grant	Permission originates at parent	Owner/manager changes origin; record blocker

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 4 Posting lifecycle and asset metadata

20 minutes · Slides 38–46

Product sheet P-101 has three competing copies. Sales sent an obsolete price. Create a single source of record with an explicit business publication lifecycle.

Goal: One current P-101 source is registered, price is SGD 49, approval checkpoint and viewer evidence exist, and obsolete reference is retired.

Tools: Drive, Docs, Sheets

Additional mechanism/chart slides: None.

Mock inputs: asset-register.csv, product-copy.md, scenario.json.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Local package: activities/activity-04-posting-lifecycle-and-asset-metadata/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
asset_id	P-101	Stable business identifier
status	Draft / Review / Approved / Published	Business register states
source_url	Working Drive item	Do not substitute obsolete copy
published_on	2026-09-13	Release evidence and next review

Business states: Draft and Approved are fields in our register, not a claim that Drive enforces a complete CMS approval workflow.

Stable source: A single source URL reduces ambiguity. Downloads and exported PDFs are snapshots, so published copies require review.

Version naming: Named editor versions make approved checkpoints inspectable; changing the filename alone does not prove content approval.

Retirement: An obsolete public file needs removal from current navigation and a controlled archive policy; record references before retirement.

Detailed procedure

1. Open asset-register.csv and product-copy.md. Identify P-101's conflicting price values.

2. Create a Restricted Google Doc for P-101 and paste the product copy. Correct the price to SGD 49.00.

3. Create or import the asset register into Sheets and add source_url, approver, published_on and review_due columns.

4. Name the current document version Draft v 1, ask a partner to comment on accuracy, and resolve the review finding.

5. Record the approval in the register and name the corrected version Approved v 2.

6. Create an approved release PDF and keep its URL separate from the editable source URL. For this exercise share only with named test identities.

7. Update the index to the approved current copy and mark the older mock copy Superseded.

8. Open the registered URL as the intended viewer and verify price, title and release note. Save the completed register and release PDF.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
META	P-101	ID, owner, status, due date populated
RELEASE	v 2	Price SGD 49 and approval note
URL	Published copy	Current intended audience
OLD	v 1 reference	No longer used in current index
Symptom	Cause	Corrective check
Sales shares older PDF	Snapshot does not update with source	Replace public reference; verify release ID
Two current source links	Duplicate records for same asset ID	Choose authorised source and archive duplicate

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Unapproved content publicly visible	Audience broadened before review	Restrict; approve and republish deliberately

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 5 Search recovery and version diagnosis

20 minutes · Slides 47–55

The warehouse manager cannot find the approved delivery checklist. The search result contains a similarly named draft, and an edit removed an important dispatch check.

Goal: Correct checklist identity is found and the approved address verification line is restored with an inspectable recovery log.

Tools: Drive, Docs

Additional mechanism/chart slides: None.

Mock inputs: dispatch-checklist.md, recovery-log.csv, scenario.json.

Local package: activities/activity-05-search-recovery-and-version-diagnosis/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Search terms	delivery checklist	Match title then verify asset ID
Type filter	Documents	Reduce irrelevant results
Location	Approved Assets	Distinguish draft from release
Version history	named checkpoint	Requires appropriate access

Identity check: A familiar filename is insufficient. Match asset ID, owner, location and current release record.

Recovery target: A version restore changes current content; compare the selected checkpoint and keep a review record.

Editor versus binary: Native editor version history and uploaded-file versions are different mechanisms. Inspect the appropriate history for the item type.

Access failure: Not found and not authorised can look similar. Verify the invited identity and known source URL before creating a duplicate.

Detailed procedure

1. Create a Google Doc from dispatch-checklist.md in your training folder and name its version Dispatch v 1.

2. Add the customer-address verification line and name the version Dispatch v 2 approved.

3. Make a later test edit that removes the line and clearly label it as a deliberate training fault.

4. Search Drive for the checklist title and apply the Documents type filter. Record the returned item identity and location.

5. Open File then Version history then See version history. Compare the named approved checkpoint with current content.

6. With owner/editor permission restore the approved checkpoint or copy the approved text back with an explicit recovery note.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

7. Ask your partner to confirm the address-check line and open the current source URL.

8. Complete recovery-log.csv with item ID, selected checkpoint, reason and final validation.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
FIND	DELIVERY-01	Correct location and identifier
HISTORY	Dispatch v 2	Approved checkpoint identified
RECOVER	Current text	Address check restored
ACCESS	Intended viewer	Known URL opens
Symptom	Cause	Corrective check
Correct file absent from search	Wrong account, location or access	Test known URL and invited identity
Draft mistaken for current	Title reused without asset identifier	Validate register and approved location
Restored content still incomplete	Wrong checkpoint selected	Compare explicit acceptance lines

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Content assets and customer web properties

LO2 · Slide range 56–200

Activity 6 Docs structure and collaborative review

25 minutes · Slides 57–65

A delivery policy is hard to navigate and reviewers overwrite each other. Build a structured document and distinguish proposed edits from accepted content.

Goal: Heading outline works; supported 14-day return correction is accepted, unsupported change rejected, and approved version evidence is saved.

Tools: Docs, Drive

Additional mechanism/chart slides: None.

Mock inputs: delivery-policy.md, policy-facts.csv, scenario.json.

Local package: activities/activity-06-docs-structure-and-collaborative-review/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Heading styles	Heading 1 / Heading 2	Structure supports outline and navigation
Mode	Suggesting	Proposal does not replace accepted text yet
Comment	specific selection + issue	Resolve after decision
Version	Approved delivery policy	Auditable release checkpoint

Structure: Heading styles encode hierarchy; simply making text large or bold does not provide the same outline structure.

Review contract: Suggestions are proposals. An authorised editor accepts or rejects them; comments record discussion and can be resolved.

Source validation: The policy must trace delivery price and return window to a business-approved source, not an invented reviewer assumption.

Release state: A named approved checkpoint plus an approval record identifies the version intended for distribution.

Detailed procedure

1. Create a Google Doc from delivery-policy.md. Rename it HHS DP-01 delivery policy followed by your initials.

2. Apply Heading 1 to Delivery eligibility and Returns; apply Heading 2 to Standard delivery.

3. Check the document outline and confirm both main sections appear.

4. Share the training document with your partner as Commenter and ask them to propose the provided correction using Suggesting or a comment if suggesting is unavailable.

5. Select the return-window text and record the verified source from policy-facts.csv. Do not accept unsupported changes.

6. As editor accept the supported correction and reject the deliberately unsupported free-delivery suggestion.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

7. Resolve the discussion only after checking final content. Name the current version Approved DP-01 v 2.

8. Export PDF and save review-evidence.md containing proposed change, decision, source and final value.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
STRUCT	Outline	Two Heading 1 sections visible
REVIEW	Suggestion	14-day proposal recorded
DECISION	Accepted text	Verified terms reflected
RELEASE	Named version	DP-01 v 2 approved
Symptom	Cause	Corrective check
Outline missing headings	Direct formatting used instead of styles	Apply Heading 1/2 to structure
Reviewer overwrites content	Editing mode used for review	Use Suggesting and review role
Comment resolved but text wrong	Discussion closure mistaken for correction	Check final policy text and approval

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 7 Docs templates and publication readiness

20 minutes · Slides 66–74

Sales reuses a brochure containing an old address and an internal cost figure. Apply a publication checklist to a controlled template before export.

Goal: Brochure has no template placeholders or internal cost, accurate product facts, a tested descriptive link, and a readable approved PDF.

Tools: Docs, Drive

Additional mechanism/chart slides: None.

Mock inputs: brochure-template.md, product-facts.csv, publication-checklist.csv, scenario.json.

Local package: activities/activity-07-docs-templates-and-publication-readiness/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Template fields	product, price, contact, version	No unreplaced placeholders
Internal fields	unit_cost	Exclude from public copy
Link text	View delivery terms	Descriptive purpose
Image text	Alternative text	Meaningful description where appropriate

Template scope: A template constrains structure but cannot prove correctness. Each copy still needs a source and reviewer.

Redaction: Deleting a visible cost from the public export is only one check. Review comments, links, embedded tables and metadata of the distributed artifact.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Readable links: Descriptive link text helps a reader predict destination; test the viewer's permission on the target as well as the link itself.

Export divergence: PDF is a snapshot. A later source edit requires a newly reviewed export and replacement of the published reference.

Detailed procedure

1. Create a Google Doc using brochure-template.md and the product-facts.csv mock data.

2. Replace all brace placeholders and set the product name, SGD 29 price and example contact.

3. Remove the internal unit_cost field and review comments for internal notes.

4. Apply heading styles and add descriptive link text for the delivery policy.

5. Insert an optional locally supplied illustration and add an accurate alternative text description through Image options or the available Alt text control.

6. Review the publication-checklist.csv row by row and record pass or corrective action.

7. Download the document as PDF, open it, and inspect every page and hyperlink with a test viewer.

8. Save the approved brochure PDF and completed release checklist.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
FACT	Price	SGD 29 matches approved product record
LEAK	unit_cost	Absent from public export
LINK	Delivery terms	Intended viewer can open
PDF	Release copy	Current version and readable layout
Symptom	Cause	Corrective check
Placeholder remains	Template fields incomplete	Search for double-brace markers
PDF contains old address	Export predates source update	Regenerate from approved current source
Link gives permission error	Target ACL differs from brochure	Share approved target to intended audience

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 8 Sheets import schema and data validation

25 minutes · Slides 75–83

Enquiry data has duplicate IDs, numeric amounts stored as text and inconsistent stages. A marketing report is unreliable until the data contract is validated.

Goal: Clean sheet contains 12 unique enquiries, valid stages and numeric amounts; duplicate/type decisions and invalid-input rejection are evidenced.

Tools: Sheets

Additional mechanism/chart slides: None.

Mock inputs: cleaning-log.csv, enquiries-dirty.csv, scenario.json, source-notes.md.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Local package: activities/activity-08-sheets-import-schema-and-data-validation/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
enquiry_id	E001...E012	Required unique identifier
amount_sgd	Numeric >=0	Currency formatting is not data cleaning
stage	New / Qualified / Won / Lost	Dropdown rejects invalid input
created_on	ISO date in source	Check interpreted date and locale

Identifier rule: Duplicate rows may be repeats rather than new customers. Resolve using the stable enquiry ID before summing revenue.

Type boundary: Formatting a text cell as currency does not guarantee numeric values. Check ISNUMBER and the source import.

Validation: An allowed-value dropdown standardises new edits; audit existing invalid values rather than assuming validation cleans them.

Locale: Dates and decimal separators depend on spreadsheet locale. ISO source dates and explicit type checks reduce ambiguity.

Detailed procedure

1. In Sheets create HHS Enquiry Data and choose File then Import then Upload to import enquiries-dirty.csv into a new sheet.

2. Rename the tab Raw and freeze the header row. Check spreadsheet locale and verify interpreted dates.

3. Duplicate the tab as Clean so the original mock input remains unchanged.

4. Add checks using COUNTIF for duplicate IDs and ISNUMBER for amount values.

5. Review duplicate E003 against source-notes.md and keep one authorised record. Record the decision in cleaning-log.csv.

6. Standardise stages to New, Qualified, Won or Lost and apply a Data validation dropdown to the stage column with invalid input rejected.

7. Format numeric amounts as SGD currency and confirm numeric values remain numbers.

8. Try a deliberate invalid stage in a spare test row, confirm rejection, remove the test row and export the cleaned CSV.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
ROWS	Input	12 unique IDs after review
STAGE	Validation	Invalid PendingRejected input blocked
TYPE	Amounts	ISNUMBER true on report rows
DUP	E003	Duplicate resolution documented
Symptom	Cause	Corrective check
Revenue unexpectedly zero	Amounts imported as text	Test ISNUMBER; correct source/import
Stage groups split	won and Won differ	Standardise against allowed stages

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Revenue inflated	Duplicate enquiry ID counted twice	Review duplicate IDs before removal

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 9 Sheets formulas and operational checks

25 minutes · Slides 84–92

Operations needs the enquiry pipeline value and conversion rate. Compute explicit numerators and denominators rather than relying on an unlabeled percentage.

Goal: Total 12, Won 3, Won revenue SGD 960, conversion 25%, average won SGD 320 all match hand calculation and report labels identify the cohort.

Tools: Sheets

Additional mechanism/chart slides: 246,247,248,249,250,253.

Mock inputs: enquiries-clean.csv, hand-check.md, products-lookup.csv, scenario.json.

Local package: activities/activity-09-sheets-formulas-and-operational-checks/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Total enquiries	=COUNTA(A2:A13)	12 validated IDs
Won enquiries	=COUNTIF(C2:C13,"Won")	3 Won records
Won revenue	=SUMIF(C2:C13,"Won",D2:D13)	SGD 960
Conversion	=IF(B2=0,"No data",B3/B2)	Summary B2 total 12; B3 Won 3; result 25%

Named denominator: A 25% conversion rate here means Won enquiries divided by all 12 validated enquiries. It does not mean website visitor conversion.

Conditional aggregation: SUMIF matches the stage column and sums the corresponding amount range. Keep both ranges aligned.

Empty period: IFERROR avoids a divide error but a displayed zero can hide no data. Display an explicit no-data label in the report.

Verification: Independent hand calculation on three won records should match the spreadsheet result before charts are published.

Detailed procedure

1. Import enquiries-clean.csv into Sheets, rename the imported tab exactly Clean and confirm 12 enquiry rows plus one header. Formula references below depend on this tab name.

2. Create a Summary tab. Put labels Total enquiries, Won enquiries, Won revenue, Conversion andAverage won value in A2: A6; put their values/formulas in B2: B6.

3. Enter =COUNTA(Clean!A2: A13) and =COUNTIF(Clean!C2: C13,"Won") in the first two value cells.

4. Enter =SUMIF(Clean!C2: C13,"Won", Clean!D2: D13) for revenue.

5. In Summary B5 enter =IF(B2=0,"No data", B3/B2) and format numeric results as percent. B2 is total enquiries and B3 is Won enquiries; label this denominator explicitly.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

6. Calculate average won value as Won revenue divided by Won count and format as SGD currency.

7. Use the provided hand-check.md to independently add 360,180 and 420 and compare against the sheet.

8. Add an empty-period test with a No data label, then save the report evidence and export the summary PDF.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
COUNT	Unique IDs	12
WON	Stage rows	3
REV	Won revenue	SGD 960
RATE	Won / enquiries	25% with cohort label
Symptom	Cause	Corrective check
SUMIF wrong total	Sum range offset from criteria range	Align row 2:13 on both ranges
Conversion exceeds 100%	Numerator and denominator from different populations	Use same validated enquiry cohort
Zero interpreted as failure	Empty period masked by IFERROR	State No data alongside result

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 10 Filter views pivots and editable charts

25 minutes · Slides 93–101

Sales and Operations need different filtered views without disrupting each other. Summarise enquiry stages and visualise counts using the validated 12-row cohort.

Goal: Filter view does not disrupt peer view; pivot counts 4/3/3/2 total 12; chart is editable and correctly labels enquiry counts.

Tools: Sheets

Additional mechanism/chart slides: 251,252.

Mock inputs: enquiries-clean.csv, scenario.json.

Local package: activities/activity-10-filter-views-pivots-and-editable-charts/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Basic filter	Shared sheet view	May change what other users see
Filter view	Saved named view	Does not disrupt another user view
Pivot rows	stage	Group New/Qualified/Won/Lost
Pivot values	COUNTA enquiry_id	Counts records, not amount totals

View versus data: A filter hides rows in a view; it does not delete them. Verify the source range before interpreting a chart.

Pivot aggregation: Count of IDs answers how many enquiries; SUM of amount answers pipeline value. These are different measures.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Chart choice: A column chart compares stage counts. A line chart would imply an ordered temporal trend that these categories do not provide.

Collaboration test: Open the sheet as a second editor and verify a filter view selection does not alter the other editor's working view.

Detailed procedure

1. Import enquiries-clean.csv and name the tab Clean. Select the full header and 12 data rows.

2. Create a named filter view for Qualified enquiries using the Data menu Filter views control available to your account.

3. Ask a paired editor to confirm their own view remains unchanged; record the distinction from a basic filter.

4. Create a pivot table from Clean A1: D13 with stage as rows and COUNTA of enquiry_id as values.

5. Check that New 4, Qualified 3, Won 3 and Lost 2 sum to 12.

6. Insert a column chart from the stage/count summary, excluding the grand-total row.

7. Set the chart title to Validated enquiry stages and label the measure Enquiry count.

8. Export the pivot summary and capture chart/source evidence; save the view and verify again as a second user.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
VIEW	Sales qualified	Partner view unaffected
PIVOT	Stage counts	4,3,3,2
TOTAL	Pivot count	12
CHART	Column chart	Counts match source
Symptom	Cause	Corrective check
Partner loses visible rows	Basic filter applied to shared view	Use a filter view instead
Pivot counts 11	Source range omitted final row	Inspect pivot source range
Won shown as revenue	Wrong measure/title pairing	Label count and currency separately

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 11 Slides assets and customer presentation

20 minutes · Slides 102–110

Marketing needs a short customer presentation containing only approved product facts and a pipeline chart with a clear data period.

Goal: Three customer slides use approved facts, clear visual hierarchy and synthetic-data chart label; private cost details are absent and viewer access is verified.

Tools: Slides, Sheets, Drive

Additional mechanism/chart slides: None.

Mock inputs: pitch-outline.md, product-facts.csv, scenario.json, stage-counts.csv.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Local package: activities/activity-11-slides-assets-and-customer-presentation/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Slide title	Storage that fits your home	Specific customer-facing message
Price field	SGD 29	Matches approved source
Chart period	September 2026 synthetic cohort	Identifies sample context
Source link	Controlled Google Sheet	Viewer access separate from edit

Visual hierarchy: Title, image and concise verified values make the decision legible; the source register still governs correctness.

Chart dependency: A linked Sheets chart is a dependency. A later update needs review before refreshing the customer deck.

Audience boundary: Sharing a presentation does not necessarily grant the viewer access to every linked source file.

Privacy: Remove internal commentary and cost values from the customer copy; inspect the exported PDF and permission state.

Detailed procedure

1. Create a new Slides presentation named HHS Customer Pitch using the local pitch-outline.md.

2. Create three slides using title and image layouts, keeping product price SGD 29 and approved delivery/return terms.

3. Insert a locally supplied product illustration or an approved mock image; add accurate alternative text.

4. Insert or link a Sheets chart from the validated stage counts and label it Synthetic September 2026 enquiry cohort.

5. Use consistent theme fonts and check the visual hierarchy in presentation mode.

6. Review speaker notes, comments and hidden content for internal margin/cost values and remove them from the customer copy.

7. Share only with your named training viewer and ask them to open the presentation.

8. Download PDF and inspect all pages. Save the source presentation URL, release PDF and publication checklist.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
PRICE	Customer slide	SGD 29
CHART	Stages	4/3/3/2 with synthetic label
PRIVATE	Notes/comments	No internal cost values
VIEW	PDF/viewer	Readable and accessible
Symptom	Cause	Corrective check
Linked chart out of date	Source changed after insertion	Review current data and refresh deliberately
Customer cannot open source	Source ACL separate	Share approved source or use snapshot

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Notes contain internal margin	Source deck reused without release review	Remove private notes and inspect export

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 12 Forms publication consent and response schema

25 minutes · Slides 111–119

Harbour Home Supplies needs a customer enquiry form. Its responder audience, email collection and response visibility are independent choices with privacy consequences.

Goal: Published responder link works; consent is required; private results summary is off; response sheet stays restricted and synthetic response is correctly captured.

Tools: Forms, Sheets

Additional mechanism/chart slides: None.

Mock inputs: enquiry-form-spec.md, scenario.json, test-response.csv.

Local package: activities/activity-12-forms-publication-consent-and-response-schema/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Question fields	name, email, product, consent	Collect minimum relevant data
Collect email	Verified or responder input	Choose deliberately and disclose
Limit 1 response	Google sign-in required	Not suitable for all anonymous visitors
Summary visibility	View results summary	May reveal full text to responders

Publish requirement: Current Forms requires publication before responders can access a form. Editor sharing and responder access are separate.

Email collection: A one-response limit requires sign-in but does not by itself record the username; email collection is a distinct setting.

Summary risk: Response summaries can expose full-text answers or charts to responders. Keep summary visibility off for private enquiries.

Response ACL: A published enquiry form does not justify publishing the linked response spreadsheet containing customer contact details.

Detailed procedure

1. Create a blank Form from enquiry-form-spec.md and give it the title HHS Product Enquiry Training.

2. Add product dropdown, contact-name, email and required consent questions using the supplied mock question text.

3. Choose email collection deliberately. For this scenario use responder input and describe its purpose in the form introduction.

4. In Settings review Limit to 1 response and the sign-in consequence. Leave it off for the visitor scenario unless your trainer sets a restricted classroom audience.

5. Turn off View results summary and explain the privacy reason in your evidence log.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

6. Link responses to a new Google Sheet and keep the response spreadsheet Restricted to authorised staff/test owner.

7. Publish the form to the authorised training responder audience and copy the responder link, not the editor link.

8. Open as a test responder, submit the provided synthetic record, inspect its row in the response sheet and verify that other responses are not exposed.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
PUB	Responder URL	Published training form opens
CONSENT	Test response	Required contact permission
SUMMARY	Responder view	No private response summary
SHEET	Response spreadsheet	Staff-only Restricted
Symptom	Cause	Corrective check
Responder sees unpublished form	Form draft not published	Publish to authorised training responders
Private answers visible	Results summary enabled	Disable summary and retest responder view
External visitor blocked	Audience or sign-in restriction	Align responder controls to intended audience

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 13 Forms validation and customer feedback analysis

25 minutes · Slides 120–128

A feedback survey contains satisfaction ratings outside 1–5 and missing responses. Validate inputs and interpret the sample without turning a small survey into a population claim.

Goal: Validated 10-response sample has mean 3.6/5 and 6/10 positive ratings; report labels synthetic scope and proposes a measurable website improvement.

Tools: Forms, Sheets

Additional mechanism/chart slides: 254.

Mock inputs: feedback-form-spec.md, feedback.csv, scenario.json.

Local package: activities/activity-13-forms-validation-and-customer-feedback-analysis/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
rating	Integer 1..5	Reject out-of-range input
issue_category	Delivery / Product / Website	Controlled choices
response_id	F001..F010	Unique synthetic record
mean_rating	=AVERAGE(B2:B11)	36/10 =3.6

Validation: The 1–5 boundary is part of the question contract; distinguish unanswered from a genuine low score.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Aggregation: The mean 3.6 uses 10 valid responses. A positive share 60% uses ratings 4 or 5 and the same 10-response denominator.

Interpretation: Synthetic convenience feedback is evidence for practising analysis, not a verified claim about real customer satisfaction.

Action link: Tie the lowest Website scores to a testable navigation improvement and define what evidence would indicate success.

Detailed procedure

1. Create or update a training feedback Form using feedback-form-spec.md with a required 1–5 rating and controlled issue category.

2. Test the input boundary by attempting an invalid value if the chosen question type permits it, or verify the fixed scale endpoints.

3. Import feedback.csv into a separate Sheets analysis tab; these are synthetic practise responses, not live survey data.

4. Check ten unique response IDs, rating bounds and missing-value count.

5. Calculate the rating mean using AVERAGE on the ten numeric values and independently add them to 36.

6. Calculate the proportion rating 4 or 5 using COUNTIF for values>=4 divided by 10.

7. Create a column chart of issue counts and a short analysis note labeling the sample synthetic.

8. Propose one website navigation test with an acceptance measure and save the feedback report and validation evidence.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
N	Valid sample	10
MEAN	Rating	3.6/5
POSITIVE	Rating 4 or 5	6/10 =60%
ACTION	Website issue	Navigation test defined
Symptom	Cause	Corrective check
Mean greater than 5	Out-of-range value included	Validate 1..5 and quarantine invalid rows
Blank counted as zero	Missing interpreted as negative feedback	Separate missing from rating 1
60% called customer-wide satisfaction	Sample overgeneralised	Label synthetic 10-response sample

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 14 Gmail routing labels and search controls

20 minutes · Slides 129–137

Sales loses customer enquiries among newsletters. Route only matching messages to a label and verify that a reply is treated according to its own criteria.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Goal: Positive and negative message tests match the declared criteria; action is label only and the report distinguishes reply criteria from conversation grouping.

Tools: Gmail

Additional mechanism/chart slides: None.

Mock inputs: filter-spec.md, mock-emails.csv, scenario.json.

Local package: activities/activity-14-gmail-routing-labels-and-search-controls/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Search	subject:(HHS enquiry)	Test matching messages first
Action	Apply label Enquiries	Label need not archive message
Existing messages	Apply filter to matching conversations	Choose explicitly
Reply behaviour	Criteria checked on message	Replies need their own match

Matching contract: A filter applies the criteria to messages. A reply without matching fields need not inherit the same filter action.

Labels: A label categorises a message without necessarily removing it from Inbox; Archive is a separate action.

Search before mutation: Preview the exact search matches before saving a routing rule, particularly before forwarding or deletion.

Privacy boundary: Automatic forwarding is a data-disclosure action. The training rule applies a local label only.

Detailed procedure

1. Read mock-emails.csv and filter-spec.md; no real customer message is needed.

2. Use trainer-provided received synthetic messages in a dedicated classroom test mailbox: two enquiry subjects and two non-matching subjects. Unsent drafts cannot demonstrate incoming-filter application. If received test mail is unavailable, document the planned test and do not claim the routing passed.

3. In Gmail search subject:(HHS enquiry) and inspect the preview matches.

4. Create an Enquiries training label, then create a filter from the search criteria.

5. Select Apply the label and choose the training label. Do not select forwarding or Delete it.

6. If you intend to label existing synthetic messages, choose the matching-existing option deliberately.

7. Check a matching message, a non-matching message and a reply whose subject differs; record observed outcomes.

8. Remove the training filter and optional label after saving evidence.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
MATCH	M01/M03	Enquiries label present
NEGATIVE	M02/M04	Label absent
RULE	Filter action	No forwarding or delete

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

CLEANUP	Training rule	Removed after evidence
Symptom	Cause	Corrective check
Newsletter labeled	Criteria too broad	Preview and narrow subject/sender
Reply not labeled	Reply fails saved criteria	Inspect its actual fields
Old mail unaffected	Existing matching option not selected	Apply explicitly only if intended

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 15 Gmail delegation and customer response governance

20 minutes · Slides 138–146

Customer-support staff need access to a shared support mailbox. Sharing the password creates an identity and accountability problem; compare delegation with forwarding or alias use.

Goal: Access design distinguishes mailbox account from alias, documents delegate scope and revocation, and produces a fact-checked customer response without shared credentials.

Tools: Gmail; delegation demonstration

Additional mechanism/chart slides: None.

Mock inputs: access-request.md, mailbox-options.csv, reply-template.md, scenario.json.

Local package: activities/activity-15-gmail-delegation-and-customer-response-governance/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Delegate	Named Google account	Mailbox permissions, not password sharing
Allowed	Read, send, delete email	Broad mailbox access
Not allowed	Change owner password	Not a full account takeover
Alias	Alternate email address	Not independent delegatable account

Delegation scope: A delegate can manage email, including deleting messages. Grant only to trusted support staff and record review ownership.

Account boundary: An alias is an alternate address, not a separate Google Account. Do not design a delegated mailbox around an alias.

Restrictions: Delegation does not grant all non-email integrations or account security controls. Verify the current surface's supported features.

Customer handling: Use an approved response template and link to current public terms, while keeping personal customer details inside the restricted mailbox.

Detailed procedure

1. Open mailbox-options.csv and classify each address as account or alias.

2. Draft a delegation request for the real support mailbox architecture using access-request.md. Use example addresses only.

3. As authorised mailbox owner in a classroom demonstration, inspect Gmail Settings then Accounts and Import then Grant access to your account if available.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

4. Do not add a live delegate without trainer-managed classroom authorisation; if unavailable, record the restriction and complete the design evidence.

5. Create a customer response draft using reply-template.md and replace product and delivery placeholders from the approved facts.

6. Check that the draft reveals no internal margin or unrelated customer details.

7. Complete a revocation/offboarding checklist naming who removes delegation and when.

8. Save the response draft and access decision showing why password sharing and alias-as-mailbox are rejected.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
ARCH	Mailbox	Account versus alias identified
SCOPE	Delegate	Read/send/delete documented
REPLY	Template	Correct terms and current links
EXIT	Procedure	Delegation-removal owner named
Symptom	Cause	Corrective check
Alias absent from delegated accounts	Alias has no separate mailbox account	Use correct mailbox architecture
Staff uses owner password	Shared credential bypasses accountability	Replace with authorised named access
Delegate retains access after exit	Delegation not revoked	Remove delegation and review grants

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 16 Calendar availability and booking boundaries

20 minutes · Slides 147–155

Sales offers consultations to Singapore customers while a partner works in another time zone. An availability link should show bookable slots without exposing private event details.

Goal: Booking route: timezone is correct, configured busy slot is excluded and a test booking creates an event. Standard-event fallback: invitation displays the same Singapore/UTC instant and intended audience; booking checks are explicitly Not executed.

Tools: Calendar

Additional mechanism/chart slides: None.

Mock inputs: calendar-events.csv, scenario.json.

Local package: activities/activity-16-calendar-availability-and-booking-boundaries/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Time zone	Asia/Singapore UTC+8	Label exported meeting times
Duration	30 minutes	Defined consultation slot
Availability	10:00–12:00 weekdays	Business policy for this scenario
Sharing level	Free/busy where appropriate	Separate from full event details

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Time contract: An absolute instant and display timezone are different. Record timezone explicitly in invitations and customer instructions.

Conflicts: Booking availability reflects configured calendars and busy events. Verify the selected conflict calendars rather than assuming all calendars are checked.

Privacy: Free/busy reveals availability without necessarily sharing event titles. Review the calendar's actual permission level.

Edition boundary: Appointment schedules and premium capabilities depend on account/subscription. A standard event invitation is the classroom fallback.

Detailed procedure

1. Create a new training calendar if allowed and set its timezone to Asia/Singapore.

2. Import or manually enter the three synthetic events from calendar-events.csv; include a 10:30–11:00 busy block.

3. Create a 30-minute appointment schedule for the stated training hours if available. Otherwise create a standard event and document the fallback.

4. For appointment schedules review the calendars included in conflict checking and booking-page audience. For the standard-event fallback review the guest list, availability/free-busy information available to you, and agree a non-conflicting test time with your partner.

5. Check calendar sharing and avoid exposing private event details to the test customer.

6. If appointment schedules are available, open the booking link as the intended test visitor and inspect timezone and available slots. If unavailable, open the standard event invitation as your invited partner and compare the displayed Singapore time and UTC instant.

7. For the booking route, book one harmless test slot and verify the resulting event and busy-state change. For the standard-event fallback, accept the invitation and verify the same instant and intended audience; do not claim booking-page or conflict-exclusion behaviour was executed.

8. Cancel the test booking or remove the standard test event. Save timezone and privacy evidence. Label booking/conflict checks Not executed when the account does not offer appointment schedules.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
TIME	10:00 +08:00	02:00 UTC same instant
BUSY	10:30–11:00	Excluded where configured
PRIVACY	Viewer	No private titles
BOOK	Training slot	One booking creates expected event
Symptom	Cause	Corrective check
Customer books busy time	Relevant calendar not checked	Review conflict-calendar selection
Meeting shifted 8 hours	Timezone omitted or misread	Compare Singapore and UTC instant
Customer sees private title	Calendar sharing too broad	Choose intended free/busy audience

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity 17 Meet collaboration and recording controls

20 minutes · Slides 156–164

A product-review meeting needs decisions and action owners. Recording and transcripts are not assumed available or appropriate for every account or meeting.

Goal: Approved tab only is shared, written decision and two owned actions persist, and recording limitations/privacy controls are accurately documented.

Tools: Meet, Calendar, Drive

Additional mechanism/chart slides: 245.

Mock inputs: decision-record.md, meeting-agenda.csv, scenario.json.

Local package: activities/activity-17-meet-collaboration-and-recording-controls/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Meeting identity	Calendar organiser	Authorised host controls
Screen share	Approved browser tab	Avoid unrelated windows
Recording	Edition/admin/role dependent	Do not assume free availability
Action item	owner, due_on, status	Business decision record

Host boundary: Meeting controls depend on host, account and administrator settings. Review who can join and present before sharing sensitive content.

Screen disclosure: Sharing a tab narrows the visible surface compared with an entire screen, but learners still check the content and notifications.

Recording scope: Availability and save location depend on the edition and meeting context. Recording also requires an approved purpose and participant notice.

Decision persistence: A written decision/action record is the durable output for this exercise; it is not replaced by an unreviewed recording.

Detailed procedure

1. Create a training Calendar event for a short paired product-review meeting.

2. Attach or link only the approved mock brochure and decision-record.md template.

3. Join Meet with your partner and inspect host controls available to your account.

4. Share the single approved product-document tab; verify with your partner what is visible.

5. Discuss the three mock agenda items and record one decision and at least two actions with owners and due dates.

6. Do not record the exercise. Inspect recording availability only in a trainer demonstration and note account/admin limitations.

7. Save the written decision record in the Restricted project folder and share it only with the intended named participants.

8. End the meeting and verify action owners can open the decision record.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
INVITE	Participants	Intended training audience

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

SHARE	Visible content	Approved product tab
DECISION	Brochure	Approval and rationale recorded
ACTION	Marketing lead	Due 2026-09-16; Open
Symptom	Cause	Corrective check
Recording control absent	Edition/admin/role limitation	Use written record; note limitation
Customer sees internal window	Wrong share surface	Stop share; select approved tab
Action has no owner	Discussion not translated to work item	Assign owner and due date

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 18 Chat spaces tasks and whiteboarding replacement

20 minutes · Slides 165–173

A product launch needs a persistent discussion space and collaborative ideas. Legacy Jamboard instructions are unusable because the application ended 31 December 2024.

Goal: Persistent discussion or documented fallback, owned task and three measurable ideas exist; no instructions depend on retired Jamboard.

Tools: Chat, Slides, Tasks if available

Additional mechanism/chart slides: 244.

Mock inputs: customer-journey.csv, idea-board.md, scenario.json.

Local package: activities/activity-18-chat-spaces-tasks-and-whiteboarding-replacement/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Space membership	Internal training users	Review external audience policy
Thread	Launch decision topic	Keep rationale attached to discussion
Task fields	assignee, due date, status	Record ownership
Whiteboard alternative	Slides brainstorm	Core exercise without third-party licence

Persistence: A space keeps shared discussion context and linked records; the underlying file still has its own sharing settings.

Task ownership: A task is actionable when an assignee and due date are explicit. Verify completion evidence, not only a chat acknowledgement.

Retired product: Jamboard is end-of-life. Rebuild its original brainstorm purpose with Slides or evaluate current third-party alternatives separately.

External boundary: External-space availability depends on administrator policy and space configuration. Keep classroom members internal unless explicitly authorised.

Detailed procedure

1. Create a training Chat space if your account permits it; otherwise document the space design and use a shared Slides board.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

2. If a Chat space is available, add only your paired training learner and give it the launch title. Otherwise share the Slides brainstorm board with the named partner and record the no-space fallback.

3. For the Chat route post the mock agenda in a thread and link the approved brochure, then verify partner file access. For the fallback place the agenda and brochure link on the shared Slides board and verify partner access there.

4. For the Chat route create a task with named assignee and due date. For the fallback record the same owned task in a shared decision/action table and mark Chat task execution Not executed.

5. Create a Slides brainstorm board using idea-board.md with three columns for idea, customer friction and acceptance measure.

6. Add at least three ideas from the supplied customer-journey.csv and assign an owner to each.

7. Record Jamboard retirement and compare optional FigJam, Lucidspark or Miro only as evaluated alternatives, not required purchases.

8. Save space/task/board evidence and remove temporary classroom membership after the exercise.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
SPACE	Members	Only intended training identities
ASSET	Brochure link	Opens for member
TASK	Terms review	Owner and due date
IDEA	Slides board	Three measurable visitor improvements
Symptom	Cause	Corrective check
Member cannot open linked brochure	File ACL differs from space	Share approved asset to intended members
Task remains unassigned	Chat message substituted for task	Assign owner and due date
Jamboard link fails	Retired application	Use Slides board; retain exported legacy evidence if available

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 19 Sites architecture navigation and draft content

30 minutes · Slides 174–182

Customers cannot find delivery terms or the enquiry form on the SME site. Design a small information architecture around actual customer tasks and independent source assets.

Goal: Four task-oriented pages with approved facts and a documented independent embed ACL exist; delivery terms are discoverable within 2 clicks in the test.

Tools: Sites, Docs, Drive

Additional mechanism/chart slides: None.

Mock inputs: product-facts.csv, scenario.json, site-copy.md, usability-log.csv.

Local package: activities/activity-19-sites-architecture-navigation-and-draft-content/. Every named input file is included in this folder.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Field or control	Scenario value	Interpretation
Page hierarchy	Home, Products, Delivery, Contact	Consistent task-based labels
Primary action	Product enquiry	One visible destination
Draft	Unpublished changes	Review before publishing
Embedded asset	Approved brochure PDF	Viewer ACL checked separately

Information architecture: The site map names pages and tasks; it is a design artifact, not a broad list of site benefits.

Content dependency: A Sites page can embed a file, but the file's permission remains independent. A visible editor embed is not proof of customer access.

Draft boundary: Editing a published site creates changes requiring publication; distinguish editor preview from current visitor content.

Task test: Evaluate a visitor's ability to find the delivery price and enquiry action, then record observed completion and friction.

Detailed procedure

1. Create a new training Google Site and use site-copy.md as the approved mock content source.

2. Create Home, Products, Delivery and Contact pages with the stated navigation labels.

3. Add approved product names/prices from product-facts.csv and a clear Product enquiry action.

4. Insert the approved brochure or document from your training Drive folder and note its independent Share setting.

5. Add delivery terms SGD 5 below SGD 80 and the approved return window.

6. Preview desktop and mobile layouts and check page hierarchy and navigation labels.

7. As site author open Preview and have your partner perform the delivery-price task on the author's displayed preview; record clicks and confusion in usability-log.csv. Alternatively explicitly invite the paired learner as an editor for draft preview. This is a draft usability test; published visitor access is verified in Activity 20.

8. Save the site draft and evidence; publish only to the authorised training audience in the next activity.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
MAP	Four pages	Task labels consistent
FACT	Delivery	SGD 5 / threshold SGD 80
EMBED	Brochure	Intended viewer access
TASK	Delivery terms	Located within 2 clicks
Symptom	Cause	Corrective check
Viewer sees permission request	Embedded file restricted to editors	Share approved release file appropriately
Terms absent from live site	Draft changes not published	Review and publish current change
Navigation uses internal jargon	Labels reflect departments, not visitor tasks	Use product/delivery/contact task labels

Primary source: Current Google guidance

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 20 Sites publishing embed access and maintenance

30 minutes · Slides 183–191

The editor sees a brochure and form correctly, but a customer gets permission errors. Validate site publication, responder access and embedded file sharing as separate controls.

Goal: Exact published site, brochure and form work for intended viewer; response sheet denies unauthorised access and maintenance checklist has resolved findings.

Tools: Sites, Drive, Forms

Additional mechanism/chart slides: None.

Mock inputs: maintenance-checklist.csv, scenario.json.

Local package: activities/activity-20-sites-publishing-embed-access-and-maintenance/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Editor URL	Sites editing surface	Not the visitor release URL
Published URL	Published training site	Test current content
Drive embed	File General access + People	Separate ACL
Form embed	Publish + responders	Separate responder contract

Three audiences: Site viewers, embedded-file viewers and form responders are separate permission contracts. Align them deliberately.

Response privacy: The form can accept enquiries while its response sheet remains staff-only. Do not grant customers access to the response store.

Maintenance cycle: Check content facts, broken links, mobile layout, audience and review dates. Record findings and corrective action for each item.

Live verification: Open the exact published URL with the intended viewer. Editor preview does not establish current visitor behaviour.

Detailed procedure

1. Open the training Sites draft from Activity 19 and review the local maintenance-checklist.csv.

2. Check approved product/delivery facts and resolve any usability finding before publication.

3. Inspect sharing on the embedded brochure and grant only the authorised training viewer audience.

4. Inspect the Form publication and responder settings; use the correct published responder link.

5. Keep the linked response spreadsheet Restricted to the owner or approved staff; do not share it to site visitors.

6. Publish the Site to the authorised training audience and copy its published visitor URL.

7. Open that exact URL with your partner’s visitor account and test navigation, brochure access and one synthetic form submission.

8. Complete every maintenance-checklist row with observation and evidence, fix failures and repeat the visitor test before saving release records.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Acceptance and troubleshooting

Evidence	Artifact	Expected result
SITE	Published URL	Current facts visible
FILE	Embed	Viewer opens brochure
FORM	Responder	Synthetic submission succeeds
PRIVATE	Response sheet	Uninvited visitor denied
Symptom	Cause	Corrective check
Site opens; brochure denied	Drive embed ACL excludes visitor	Share approved file with intended audience
Form not accepting visitor	Form unpublished or responder restricted	Publish/align responder audience
Response data exposed	Response sheet broadened with site	Restrict response store separately

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 21 CMS market offerings and weighted selection

25 minutes · Slides 192–200

An SME choosing between Google Sites, WordPress and SharePoint needs a documented content-management decision, including audience, editing workflow, integrations and operational ownership.

Goal: Comparison covers Google Sites plus 2 offerings with dated primary sources, weights 100, correct 3.65/3.90/3.45 scores and a mandatory-constraint decision.

Tools: Sheets, browser research

Additional mechanism/chart slides: 242,243,255.

Mock inputs: cms-requirements.md, offering-evidence.csv, scenario.json, selection-matrix.csv.

Local package: activities/activity-21-cms-market-offerings-and-weighted-selection/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Criteria weights	Ease 30, Public 25, Governance 25, Extension 20	Sum 100 in scenario
Score scale	1..5	Learner analysis, not vendor benchmark
Google Sites	Simple team/site authoring	Assess against required workflow
Other offerings	WordPress / SharePoint	Verify roles and deployment context

Scenario scores: The scores are an explicit fictional analyst judgement. They are not current market statistics or universal product rankings.

CMS boundary: Google Sites is a site builder within Workspace; Docs/Drive manage collaboration assets. Do not imply it supplies every workflow of a dedicated CMS.

WordPress roles: WordPress provides role/capability controls; hosting, updates and plugins introduce operational responsibilities that must be assigned.

SharePoint permissions: SharePoint sharing and modern site permissions support Microsoft 365 collaboration; validate the audience and tenant configuration for the actual use case.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Detailed procedure

1. Read cms-requirements.md and mark mandatory versus weighted criteria.

2. Open the official Google Sites, WordPress roles and SharePoint permissions documentation listed in sources.md.

3. Record the supported role/sharing/publishing contracts and research date in offering-evidence.csv.

4. Import selection-matrix.csv into Sheets and verify weights sum to 100.

5. Calculate each weighted score using SUMPRODUCT of weights and scenario scores divided by 100.

6. Confirm Sites 3.65, WordPress 3.90 and SharePoint 3.45 for the supplied fictional scores.

7. Change one requirement or weight and explain how the recommendation changes; apply mandatory constraints before interpreting the rank.

8. Save a comparison report with sources, scenario assumptions, risks, operational owner and recommendation.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
WEIGHTS	Matrix	30+25+25+20=100
MATH	Sites score	3.65
SOURCES	Offerings	Official links/date documented
GATE	Must-have	Failure excludes unsuitable option
Symptom	Cause	Corrective check
Weights total 110	Scoring arithmetic invalid	Normalise weights to 100
Vendor brochure treated as proof	Feature claim lacks contract source	Read official roles/permission documentation
High score hides unmet hard constraint	Weighted average masks mandatory need	Apply must-have gate before ranking

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Metrics add-ons and integration decisions

LO3 · Slide range 201–237

Activity 22 Workspace metrics and evidence provenance

30 minutes · Slides 202–210

The business requests document access frequency, collaboration rate and engagement. Report defined measures from labelled synthetic events without claiming unavailable tenant telemetry.

Goal: Report correctly shows 60 view events and 50% defined collaboration across 4 edited docs, with explicit synthetic source, population and limitation notes.

Tools: Sheets; Admin reporting demonstration

Additional mechanism/chart slides: 256,257,259.

Mock inputs: document-events.csv, document-summary.csv, metric-dictionary.md, scenario.json, weekly-views.csv.

Local package: activities/activity-22-workspace-metrics-and-evidence-provenance/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
View events	action=view per document	Event count is not unique visitors
Unique viewers	distinct actor_id for views	Synthetic stable identifiers
Collaboration rate	docs with>=2 editors / edited docs	Defined business measure
Data provenance	Synthetic mock / live admin report	State source and availability

Metric definition: Define event, population and denominator. View frequency counts events; unique viewers deduplicate identities; these answer different questions.

Provenance: Core activity uses synthetic event data. Live Admin reports require authorised roles and feature availability; they are a separately demonstrated source.

Collaboration rule: Our 50% rate means two of four edited documents have at least two distinct editors. It is a business-defined measure, not a guaranteed built-in report label.

Interpretation: High access is not automatically success. Pair a measurable user task or enquiry outcome with content activity and note source limitations.

Detailed procedure

1. Import document-events.csv and document-summary.csv into Sheets as separate Raw Events and Summary tabs.

2. Read metric-dictionary.md and identify action, actor_id, document_id and event date.

3. Count view events per document using a pivot table or COUNTIFS and confirm 24,18,12 and 6.

4. In a new Editor Counts tab enter DOC01–DOC04 in A2: A5. In B2 enter =COUNTUNIQUE(FILTER('Raw Events'!D$2: D$68,'Raw Events'!C$2: C$68=A2,'Raw Events'!E$2: E$68="edit")) and fill down. This independently deduplicates actor IDs from the raw event rows. Verify 3,2,1,1 against the supplied summary.

5. Flag each edited document with>=2 distinct editors and compute collaborative documents divided by all 4 edited documents.

6. Confirm total 60 view events and 50% collaboration rate with a hand check.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

7. Create a labelled monthly report stating synthetic source, September period, definitions, limitations and one corrective action.

8. If a trainer demonstrates Admin Reports, record the actual role/edition/source and differences without replacing synthetic results with invented telemetry.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
VIEWS	DOC01..04	24+18+12+6=60
EDITORS	DOC01..04	3,2,1,1
COLLAB	>=2 editors	2/4=50%
SOURCE	Report	Synthetic explicitly stated
Symptom	Cause	Corrective check
Views called customers	Repeated/internal events misclassified	Label view events and source population
Collaboration denominator wrong	All assets included instead of edited docs	Use the defined 4 edited-doc cohort
Admin metric unavailable	Role/edition/report limitation	Use labelled synthetic data; record gap

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 23 Visitor journey funnel and improvement tests

25 minutes · Slides 211–219

Marketing wants three customer-experience improvements. Use a synthetic visitor funnel and test definitions rather than asserting website analytics is automatically available from Drive.

Goal: Funnel and ratios are correct and labelled synthetic; three ideas name friction, owner, measure, threshold and test period.

Tools: Sheets, Sites

Additional mechanism/chart slides: 258.

Mock inputs: experiment-template.csv, journey-friction.csv, scenario.json, visitor-funnel.csv.

Local package: activities/activity-23-visitor-journey-funnel-and-improvement-tests/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Product reach	600/1000	60% of landing cohort
Start rate	120/600	20% of product cohort
Completion rate	72/120	60% of starters
Visitor conversion	72/1000	7.2% of landing cohort

Population alignment: Use a cohort/time period where stages are comparable. The supplied funnel is explicitly synthetic and simplified for practice.

Different denominators: Completion 60% and visitor conversion 7.2% can both be correct because their denominators differ.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Instrumentation: Real visitor measurement requires configured analytics and approved data collection. Drive file activity is not a substitute for site visitor sessions.

Testable ideas: Improve product navigation, shorten enquiry fields and offer booking. Each idea names friction, owner, metric and an acceptance threshold.

Detailed procedure

1. Import visitor-funnel.csv into Sheets and read journey-friction.csv.

2. Calculate product reach 600/1000, start rate 120/600, completion 72/120 and visitor conversion 72/1000.

3. Label all four denominators and verify 60%,20%,60% and 7.2%.

4. Create a column or funnel-style comparison chart that displays the four synthetic stage counts.

5. Write three improvements: task-based product navigation, a shorter consented enquiry form, and a consultation-booking link.

6. For each idea name customer friction, change, owner, success metric, target and test period using experiment-template.csv.

7. State that real site visitor measurement requires configured analytics and approved collection; do not claim these are live Workspace metrics.

8. Save the monthly visitor report and three measurable idea proposals.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
FUNNEL	Stages	1000/600/120/72
MATH	Conversion	7.2% landing cohort
IDEA1	Navigation	Product reach target>=70%
IDEA2/3	Form/booking	Completion>=75%; availability test
Symptom	Cause	Corrective check
Conversion 60% shown as visitor rate	Starter denominator mistaken for landing cohort	Label each ratio explicitly
Analytics reports no visitors	Instrumentation not configured/available	Verify actual collection; avoid invented data
Three ideas cannot be tested	No measure or owner	Add threshold, period and accountable owner

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 24 Add-on selection scopes and administrator controls

25 minutes · Slides 220–228

An add-on promises PDF merge and customer follow-up, but requests broad file and mail access. Recommend a suitable extension only after fit, permissions, data handling and exit checks.

Goal: One current add-on is evaluated with actual dated listing/access evidence, fit and data handling, administrator approval route, pilot checks and revocation/exit plan.

Tools: Marketplace, Sheets; administrator review

Additional mechanism/chart slides: None.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Mock inputs: addon-candidates.csv, addon-review.md, scenario.json.

Local package: activities/activity-24-add-on-selection-scopes-and-administrator-controls/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Functional fit	Merge approved PDFs	Requirement-specific evidence
Requested access	Files / mail as declared	Review actual consent wording
Admin status	Allowed / blocked / limited	Policy can prevent installation
Exit	Revoke OAuth and uninstall	Separate access and installation review

Scope boundary: A requested OAuth permission describes data access, not a quality endorsement. Compare it with the actual business requirement.

Administrator controls: Workspace app availability may be controlled by administrator policy. A learner should submit a recommendation rather than bypass a blocked app.

Data handling: Review where content is processed, retention, deletion and subcontractors using current publisher evidence. Mark unknowns explicitly.

Pilot and rollback: Use synthetic files, define output checks, limit the pilot audience and document revocation/uninstallation and output portability.

Detailed procedure

1. Open addon-candidates.csv; supplied names are fictional option profiles, not endorsed live products.

2. Search current Workspace Marketplace for one PDF/content utility relevant to the requirement, without installing it.

3. Record listing URL, publisher, research date, features and the actual requested-access wording in addon-review.md.

4. Read publisher privacy/retention/deletion evidence and mark unsupported details Unknown.

5. Compare functional fit, permissions, administrator policy, cost and exit against the supplied requirement.

6. Create an approval request for a limited synthetic-file pilot; do not bypass an administrator block or grant unnecessary scopes.

7. Define expected pilot output, failure checks and a rollback plan including revocation and uninstall review.

8. Save the recommendation with reasons to adopt, reject or defer and the remaining evidence gaps.

Acceptance and troubleshooting

Evidence	Artifact	Expected result
FIT	Candidate A	PDF requirement matched
SCOPE	Consent	Actual declared access recorded
PILOT	Mock PDFs	Correct readable merged output
EXIT	Connection/install	Revocation and uninstall procedure
Symptom	Cause	Corrective check

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Install blocked	Administrator allowlist/policy	Submit review request; do not bypass
Broad mail scope for PDF task	Permission exceeds requirement	Choose lower-access option or reject
Uninstall assumed revoke	OAuth connection may still require review	Revoke access and confirm installed state

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Activity 25 Apps Script and API integration contract

30 minutes · Slides 229–237

The SME wants update timestamps on a content register and considers automated email. A safe bound-sheet example exposes the simple-trigger boundary before evaluating authorised integration.

Goal: User edit C2 creates E2 timestamp; header/unrelated edits are ignored; report correctly states simple-trigger auth/runtime and programmatic-edit boundaries.

Tools: Sheets, Apps Script; optional developer extension

Additional mechanism/chart slides: None.

Mock inputs: automation-register.csv, request-schema.json, scenario.json, timestamp.gs, trigger-boundary.md.

Local package: activities/activity-25-apps-script-and-api-integration-contract/. Every named input file is included in this folder.

Field or control	Scenario value	Interpretation
Event object	e.range	Exists when user edit fires trigger
Guard	sheet name + column 3 + row>1	Ignore unrelated/header edits
Write	column 5 new Date()	Only bound sheet for simple trigger
Authorization	Simple trigger cannot send mail	Installable runs creator if used

Trigger identity: onEdit receives an event for a user edit; clicking Run manually has no normal edit event, so the guard returns safely.

Simple-trigger limit: Simple triggers cannot call services that require authorisation, such as sending email; their runtime limit is 30 seconds.

Programmatic edit: A script/API setValue does not cause onEdit to run. Automation needs explicit timestamp logic or another deliberately designed trigger.

Integration boundary: An installable trigger runs as its creator and requires relevant consent. A REST API client requires its own OAuth/API configuration; do not paste credentials into files.

Detailed procedure

1. Create a Google Sheet named HHS Content Register with a tab named exactly Content Register and headers asset_id, title, status, owner, updated_at.

2. Enter three mock rows from automation-register.csv in rows 2–4.

3. Open Extensions then Apps Script and replace the editor contents with the supplied timestamp.gs code.

4. Save the script. Do not create any installable trigger or grant mail access for this core activity.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

5. Return to Sheets and manually edit the status cell C2. Confirm a timestamp appears in E2.

6. Edit a header or column B and confirm the guarded script does not update an unrelated timestamp.

7. Read trigger-boundary.md and explain why manually running onEdit or a programmatic setValue is not the same as a user edit.

8. Save the script source and a test log. For the optional API design inspect request-schema.json and document OAuth/API requirements without creating credentials or calling production APIs.

```
/** @OnlyCurrentDoc */
function onEdit(e) {
  if (!e || !e.range) return;
  const r=e.range, s=r.getSheet();
  if (s.getName()!=="Content Register") return;
  if (r.getColumn()!==3 || r.getRow()<2) return;
  if(r.getNumRows()!==1 || r.getNumColumns()!==1)return;
  s.getRange(r.getRow(),5).setValue(new Date());
}
```

Acceptance and troubleshooting

Evidence	Artifact	Expected result
EDIT	Row 2 column C	Timestamp appears in E2
HEADER	Row 1	No timestamp change
OTHER	Different column	No timestamp change
SCRIPT	Programmatic setValue	No implicit onEdit event
Symptom	Cause	Corrective check
Manual Run changes nothing	No edit event object	Edit the training sheet column C
Bulk paste not stamped	Guard excludes multi-cell edits	Use single-cell task or design bulk handling
Script edit not stamped	Programmatic edit does not fire onEdit	Write timestamp explicitly in that automation

Primary source: Current Google guidance

Verified 13 September 2026. Save your completed output and exact observed evidence privately; remove temporary grants or test configuration after validation.

Supplementary mechanisms and legacy coverage

Domain verification and mail routing

Activity 1; slide 238. This extension preserves approved legacy coverage without adding mandatory activity folders.

DOMAIN SETUP CHANGE RECORD

domain: harbour.example (fictional)

TXT proof: Google-provided verification token

MX purpose: route inbound mail to Google

MX target: use current Setup Wizard record

new standard setup: smtp.google.com

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

verification status: separate from mail flow

evidence: verified domain + received test mail

Separate mechanisms: Domain verification proves control of the domain. MX records route inbound email; one does not substitute for the other.

DNS owner: Only the authorised DNS administrator changes domain records. Use the exact current Google setup instructions for the tenant.

Mail cutover: Preserve the current routing design until the approved migration cutover. Mixed old/new records require deliberate review.

Verification: Administrator evidence includes domain verified and successful inbound/outbound test messages, not merely a screenshot of DNS fields.

Detailed practice:1. Read the supplementary contract. 2. Complete the administrator or workflow change-request fields. 3. Identify the owning account/role and current policy. 4. List the authorised implementation actions. 5. Specify the exact allow/deny or output evidence. 6. Have your trainer demonstrate only in an approved tenant or record the design fallback. 7. Save the supported result and mark unexecuted checksNot executed.

Primary source: Domain verification and mail routing

User alias group and access contract

Activity 1; slide 239. This extension preserves approved legacy coverage without adding mandatory activity folders.

IDENTITY SCHEMA

user: individual Google Account with sign-in

alias: alternate address for existing account

group: membership-based mail/access audience

licence: product/service entitlement

status: active/suspended lifecycle

asset owner: individual or organisation

review: account, group, direct grants

User: A user account has a sign-in identity and service entitlements; do not treat an alias as a separately licensed login.

Alias: An alternate address can receive mail for its underlying account, but is not an independent delegated mailbox.

Group: Membership grants audience-based access; offboarding must review groups as well as direct file permissions.

Evidence: Capture the requested identity type, licence/status and expected audience. An authorised administrator implements the request.

Detailed practice:1. Read the supplementary contract. 2. Complete the administrator or workflow change-request fields. 3. Identify the owning account/role and current policy. 4. List the authorised implementation actions. 5. Specify the exact allow/deny or output evidence. 6. Have your trainer demonstrate only in an approved tenant or record the design fallback. 7. Save the supported result and mark unexecuted checksNot executed.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Primary source: User alias group and access contract

Suspension handover and deletion gate

Activity 1; slide 240. This extension preserves approved legacy coverage without adding mandatory activity folders.

OFFBOARDING CHANGE REQUEST

identity: departing Marketing editor

containment: approved suspension

inventory: My Drive files andownership

handover: transfer eligible business assets

shared drive: organisation content remains

delegation/group grants: review separately

deletion: only after approved retention/handover

Suspension: Suspension restricts account access while retaining the account's data; it is not the same as deletion or ownership handover.

Handover: Inventory individually owned assets and verify transfer using the actual administrator workflow before approving deletion.

Independent grants: Remove obsolete delegation and group membership explicitly. The lifecycle request lists each surface and its responsible owner.

Acceptance: Approved successor opens required assets; retained shared-drive content remains organisation-owned; retention decisions are documented.

Detailed practice:1. Read the supplementary contract. 2. Complete the administrator or workflow change-request fields. 3. Identify the owning account/role and current policy. 4. List the authorised implementation actions. 5. Specify the exact allow/deny or output evidence. 6. Have your trainer demonstrate only in an approved tenant or record the design fallback. 7. Save the supported result and mark unexecuted checksNot executed.

Primary source: Suspension handover and deletion gate

Service availability and account security controls

Activity 1; slide 241. This extension preserves approved legacy coverage without adding mandatory activity folders.

ADMINISTRATOR CONTROL REQUEST

OU/group: Marketing training cohort

service: Sites enabled only if approved

app policy: allowed Marketplace utilities

2-Step Verification: enrolment and policy review

licence: required edition capability

external sharing: approved audience boundary

evidence: actual policy + test identity outcome

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Policy surface: Service availability, app approval and sharing are different settings. Identify the actual policy surface and cohort.

Security: 2-Step Verification changes sign-in controls; it does not by itself restrict an already shared public file.

Availability: A missing feature may be licence, role or administrator policy. Record which evidence would distinguish these causes.

Authorisation: Core learners write a change request and test plan. Only an authorised administrator changes tenant policy.

Detailed practice:1. Read the supplementary contract. 2. Complete the administrator or workflow change-request fields. 3. Identify the owning account/role and current policy. 4. List the authorised implementation actions. 5. Specify the exact allow/deny or output evidence. 6. Have your trainer demonstrate only in an approved tenant or record the design fallback. 7. Save the supported result and mark unexecuted checksNot executed.

Primary source: Service availability and account security controls

Workspace content tools and boundaries

Activity 21; slide 242. This extension preserves approved legacy coverage without adding mandatory activity folders.

CONTENT NEED -> WORKSPACE OFFERING

Drive: store and share controlled assets

Docs: structured text and reviewed edits

Sheets: typed registers, formulas and charts

Slides: approved visual presentation

Forms: published collection and responses

Sites: site pages andembedded assets

Gmail: mail routing andcustomer reply

Content assets: Drive stores assets; Docs, Sheets andSlides author distinct artifact types. Use the appropriate source of record.

Collection: Forms publishes questions and stores responses; its responder audience differs from the response-sheet audience.

Web property: Sites assembles pages and embeds, but does not automatically broaden access to embedded files.

Selection: Awareness of market offerings includes matching Workspace offerings to a need, then comparing dedicated CMS alternatives when required.

Detailed practice:1. Read the supplementary contract. 2. Complete the administrator or workflow change-request fields. 3. Identify the owning account/role and current policy. 4. List the authorised implementation actions. 5. Specify the exact allow/deny or output evidence. 6. Have your trainer demonstrate only in an approved tenant or record the design fallback. 7. Save the supported result and mark unexecuted checksNot executed.

Primary source: Workspace content tools and boundaries

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Communication and optional extension selection

Activity 21; slide 243. This extension preserves approved legacy coverage without adding mandatory activity folders.

WORKFLOW NEED -> OFFERING

Calendar: availability and event invitations

Meet: hosted meeting and screen-share

Chat: persistent team space and discussion

Tasks: owned follow-up work where available

Apps Script: bound or authorised automation

REST APIs: OAuth-enabled client integration

Marketplace: add-on with publisher/scope review

Persistent record: Meeting/chat communication links to controlled source assets; it does not replace document ownership or publication review.

Optional capabilities: Edition/admin policy can limit booking, recording or app installation. Core exercise fallbacks are documented.

Integration: Apps Script identity and OAuth clients have explicit permission contracts; do not assume every extension runs with the same identity.

Fit: Recommend the least sufficient tool for the required artifact, privacy boundary and operating owner.

Detailed practice:1. Read the supplementary contract. 2. Complete the administrator or workflow change-request fields. 3. Identify the owning account/role and current policy. 4. List the authorised implementation actions. 5. Specify the exact allow/deny or output evidence. 6. Have your trainer demonstrate only in an approved tenant or record the design fallback. 7. Save the supported result and mark unexecuted checksNot executed.

Primary source: Communication and optional extension selection

Retired applications and current replacements

Activity 18; slide 244. This extension preserves approved legacy coverage without adding mandatory activity folders.

LEGACY UPDATE RECORD

Jamboard app: end of life 31 December 2024

Core brainstorm: shared Slides board

Currents: unavailable from 5 July 2023

Core community: Google Chat spaces

Duet AI: now Gemini for Google Workspace

AI availability: current account/admin check

AI output: source-check before publication

Jamboard: The original brainstorm learning purpose remains; procedures use a current shared Slides board instead of retired Jamboard.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Currents: Replace legacy Currents community links with a current Chat-space design and review the audience/file boundaries.

Gemini name: Duet AI for Workspace is now Gemini for Workspace. Do not preserve retired labels as current controls or assume every AI capability is enabled.

Human review: Generated text or images require source, privacy and accuracy review before becoming an approved customer asset.

Detailed practice:1. Read the supplementary contract. 2. Complete the administrator or workflow change-request fields. 3. Identify the owning account/role and current policy. 4. List the authorised implementation actions. 5. Specify the exact allow/deny or output evidence. 6. Have your trainer demonstrate only in an approved tenant or record the design fallback. 7. Save the supported result and mark unexecuted checksNot executed.

Primary source: Retired applications and current replacements

Meet Companion mode and echo prevention

Activity 17; slide 245. This extension preserves approved legacy coverage without adding mandatory activity folders.

HYBRID MEETING TOPOLOGY

meeting room hardware: room audio/video

laptop Companion Mode: second screen

laptop microphone: off

meetingaudio on laptop: off

participation: chat/presentation features

speaker/listen route: room hardware

acceptance: no feedback +intended content

Audio boundary: Companion Mode disables the laptop microphone and meeting audio to avoid echo while using room audio.

Second screen: The laptop can participate in supported collaboration surfaces while room hardware carries the conversation.

Availability: Supported features and devices vary. Verify the actual current account and device surface in the trainer demonstration.

Fault isolation: No laptop meeting audio in Companion Mode is expected. Check room-audio route rather than treating it as broken laptop speakers.

Detailed practice:1. Read the supplementary contract. 2. Complete the administrator or workflow change-request fields. 3. Identify the owning account/role and current policy. 4. List the authorised implementation actions. 5. Specify the exact allow/deny or output evidence. 6. Have your trainer demonstrate only in an approved tenant or record the design fallback. 7. Save the supported result and mark unexecuted checksNot executed.

Primary source: Meet Companion mode and echo prevention

Relative absolute and named references

Activity 9; slide 246. This extension preserves approved legacy coverage without adding mandatory activity folders.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Formula Checks!H1 tax rate 0.09

Formula Checks!D2 net value 120; D3 net value 240

=D2*(1+$H$1) ->130.8

Fill down: D3 changes; H1 stays fixed

Tax_Rate refers to 'Formula Checks'!$H$1

=D2*(1+Tax_Rate) ->130.8

Relative D2 moves when filled

Absolute $H$1 anchors the rate

Relative: A relative input reference follows the row when copied; an unanchored rate reference can drift into blank cells.

Absolute: Dollar markers lock column and row. Inspect the filled formula rather than checking only the first result.

Named range: Tax_Rate gives the rate a business meaning but still needs an accurately defined source range.

Verification: For net 120 at 9% the total 130.8; for net 240 the total 261.6. Both use the same 0.09 rate.

Detailed practice:1. Open the supplied Clean enquiry sheet. 2. Create a Formula Checks tab. 3. Enter each displayed expression and verify the listed result. 4. For absolute/named references add H1=0.09 and define Tax_Rate through Data then Named ranges. 5. For lookup import products-lookup.csv as Products. 6. For slicers use Data then Add a slicer and select the source; inspect chart/pivot and a separate SUMformula. 7. Save the expression, result and independentraw-rowcheck in your evidence log.

Primary source: Relative absolute and named references

Rounding and conditional averages

Activity 9; slide 247. This extension preserves approved legacy coverage without adding mandatory activity folders.

Clean!D2: D13 contains 12 amounts

=ROUND(2820/12,2) ->235.00

=AVERAGE(Clean!D2:D13) ->235

=AVERAGEIF(Clean!C2:C13,"Won",Clean!D2:D13)

Won amounts:360,180,420

Won average:960/3=320

display 2 decimals is not data rounding

round at defined business stage

Conditional mean: AVERAGEIF uses the qualifying stage cohort and corresponding amount range; all-enquiry mean 235 differs from Won mean 320.

Rounding: ROUND changes the computed value at a specified precision. Formatting only controls how the number is displayed.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Precision policy: Specify when to round: per line or after aggregation. Different rounding sequences can produce different financial totals.

Acceptance: Recompute sum/count for the Won cohort; verify source types and aligned ranges before trusting the mean.

Detailed practice:1. Open the supplied Clean enquiry sheet. 2. Create a Formula Checks tab. 3. Enter each displayed expression and verify the listed result. 4. For absolute/named references add H1=0.09 and define Tax_Rate through Data then Named ranges. 5. For lookup import products-lookup.csv as Products. 6. For slicers use Data then Add a slicer and select the source; inspect chart/pivot and a separate SUMformula. 7. Save the expression, result and independentraw-rowcheck in your evidence log.

Primary source: Rounding and conditional averages

Extremes ranking and ties

Activity 9; slide 248. This extension preserves approved legacy coverage without adding mandatory activity folders.

Validated amounts:120,240,360,180,90,150,300,420,60,200,500,200

=LARGE(Clean!D2:D13,2) ->420

=SMALL(Clean!D2:D13,2) ->90

=RANK(420,Clean!D2:D13,0) ->2

maximum 500; minimum 60

values 200 and 200 form a tie

top n is not the same as unique top values

rank ties need explicit business rule

Nth value: LARGE returns the nth largest value; SMALL returns the nth smallest. Repeated values participate in the order.

Rank order: RANK with 0 ranks descending; verify the chosen order and population before interpreting a league table.

Ties: Equal values can share a rank. State the tie-handling rule rather than implying a unique winner.

Verification: Sorted values independently show 500 then 420; smallest 60 then 90. These cross-check formula outputs.

Detailed practice:1. Open the supplied Clean enquiry sheet. 2. Create a Formula Checks tab. 3. Enter each displayed expression and verify the listed result. 4. For absolute/named references add H1=0.09 and define Tax_Rate through Data then Named ranges. 5. For lookup import products-lookup.csv as Products. 6. For slicers use Data then Add a slicer and select the source; inspect chart/pivot and a separate SUMformula. 7. Save the expression, result and independentraw-rowcheck in your evidence log.

Primary source: Extremes ranking and ties

Conditional minimum maximum and no matches

Activity 9; slide 249. This extension preserves approved legacy coverage without adding mandatory activity folders.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Won cohort:360,180,420

=MINIFS(Clean!D2:D13,Clean!C2:C13,"Won") ->180

=MAXIFS(Clean!D2:D13,Clean!C2:C13,"Won") ->420

criteria range and value range must match size

test cohort Absent has 0 matches

MINIFS withno match returns 0

precheck COUNTIF forNo data

zero must not be mistaken forreal minimum

Conditional extrema: The minimum and maximum apply only to the matching Won cohort, not to all enquiry amounts.

Range contract: All value and criterion ranges need equal dimensions; misalignment is a contract defect.

No matches: MINIFS returns 0 when no criterion matches; check count before presenting 0 as a real business minimum.

Acceptance: Won minimum 180 and maximum 420 agree with three raw Won rows; an absentcohort displaysNo data.

Detailed practice:1. Open the supplied Clean enquiry sheet. 2. Create a Formula Checks tab. 3. Enter each displayed expression and verify the listed result. 4. For absolute/named references add H1=0.09 and define Tax_Rate through Data then Named ranges. 5. For lookup import products-lookup.csv as Products. 6. For slicers use Data then Add a slicer and select the source; inspect chart/pivot and a separate SUMformula. 7. Save the expression, result and independentraw-rowcheck in your evidence log.

Primary source: Conditional minimum maximum and no matches

Exact match lookup and source identity

Activity 9; slide 250. This extension preserves approved legacy coverage without adding mandatory activity folders.

Products!A2: C4: ID, name, price

P101, Desk lamp,49

P102, Storage box,29

P103, Wall shelf,79

=VLOOKUP("P102",Products!$A$2:$C$4,3,FALSE) ->29

first column must contain search key

FALSE requests exact match

missing P999 ->#N/A; review source

Lookup contract: The search key must be in the first column of the range; column index 3 returns price from this three-column range.

Exact match: FALSE requests exact matching. Omitting this argument defaults to approximate mode and can yield an inappropriate result.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Stable range: Anchor the lookup range before filling rows so each enquiry searches the same approved product source.

Failure check: A missing ID should produce a visible missing-keyfinding or deliberate IFNA message, not a guessed product price.

Detailed practice:1. Open the supplied Clean enquiry sheet. 2. Create a Formula Checks tab. 3. Enter each displayed expression and verify the listed result. 4. For absolute/named references add H1=0.09 and define Tax_Rate through Data then Named ranges. 5. For lookup import products-lookup.csv as Products. 6. For slicers use Data then Add a slicer and select the source; inspect chart/pivot and a separate SUMformula. 7. Save the expression, result and independentraw-rowcheck in your evidence log.

Primary source: Exact match lookup and source identity

Slicers filters and formula visibility

Activity 10; slide 251. This extension preserves approved legacy coverage without adding mandatory activity folders.

SOURCE: Clean enquirytable

basicfilter: shared table view

filterview: saved independent view

slicer: controls matching charts/pivots

formula SUM overrange: own computation

slicer does not inherently rewrite formulas

verifychartsource andselected stage

unfiltered counts 4,3,3,2

Different surfaces: Basic filters, filter views and slicers change different presentation/analysis surfaces; choose based on collaboration need.

Slicer boundary: A slicer can filter charts/pivot tables sharing the intended dataset. Do not assume a normal SUM formula changes with the slicer.

Source contract: Charts and pivots must point to the expected data range and measure. Verify the filtered output with raw matching rows.

Practice bridge: Use the same 12 validated enquiries in Activity 10’s supplementary check; save filter selection and observed outputs.

Detailed practice:1. Open the supplied Clean enquiry sheet. 2. Create a Formula Checks tab. 3. Enter each displayed expression and verify the listed result. 4. For absolute/named references add H1=0.09 and define Tax_Rate through Data then Named ranges. 5. For lookup import products-lookup.csv as Products. 6. For slicers use Data then Add a slicer and select the source; inspect chart/pivot and a separate SUMformula. 7. Save the expression, result and independentraw-rowcheck in your evidence log.

Primary source: Slicers filters and formula visibility

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Quick reference

Expression or control	Purpose
General access / People with access	Check all audience paths separately
Heading 1 / Heading 2	Encode document structure
COUNTIF / SUMIF	Count stages and aggregate matching numeric values
SUMPRODUCT	Apply declared decision weights
Published responder link	Share the Form responder surface
Sites published URL	Test actual intended visitor view
onEdit(e)	Guarded user-edit timestamp in bound sheet

Support

Open LMS courseware portal

Support: enquiry@tertiaryinfotech.com ·+65 6100 0613 ·www.tertiarycourses.com.sg. Use mock data and keep private account evidence outside public activity repositories.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg